

Network and Systems Administrator Capstone Project

Final Project

Infrastructure Design for Vantage IT Solutions

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OVERVIEW

This document reports on the complete IT infrastructure solutions for Vantage IT Solutions, a small fictional organization. This project is written for the purpose of fulfilling the final practical assessment for the Network and Systems Administrator Program.

1. Organization and Requirements

1.1 Company Overview

Vantage IT Solutions is a small fictional start up IT solution company. The main objective of the company is to provide practical solutions, small office network deployments, network administration, and IT consulting for local small businesses. There are two locations that the company uses, a headquarters office for administration and centralized IT services, and a small branch location that provides local support for the clients. The company comprises less than 30 staff for both headquarters and its branch location.

1.2 Department and User-groups

Department	Users	Site	Access Need
Management	2	HQ	Sensitive management document
Sales	4	HQ/Branch	Clients' relationship management
Projects	5	HQ	Network and system deployment, project files
Service & Support	5	HQ/Branch	Remote Assistance, ticketing, remote support
Finance/HR/Administration	3	HQ	Administrative documents, HR records, payroll
Branch Coordination	1	Branch	Branch operation and local operations
IT administration	2	HQ	Infrastructure Administration

1.3 Organizations Structure

Level	Roles
Executive	General Manager
Business	Sales & Client Services; Admin/Finance/HR
Operations	Project lead; Service support lead; Branch Coordinator
Technology	IT administrator & assistant
Users	Staff members

1.4 Requirements

- All staff and personnel must be authenticated on one centralized server
- Authorized staff for both sites may access shared files, internal email and web application, and the DNS services
- Branch users must be securely accessing approved headquarters services
- IT personnel have a reliable inventory system, resource monitoring, a secure remote administration, and a log system
- Data configurations and project documentation must have a repeatable backup and recovery procedure

1.5 Technical Requirements

Area	Decision / Requirement
Identity	Microsoft ADDS at HQ; standard/admin accounts; workstations domain joined
DNS	AD-integrated internal DNS on HQ-DC01; branch clients query HQ DNS across VPN
DHCP	Central Microsoft DHCP; HQ FortiSwitch relay and branch Cisco IP helper-address
Routing	HQ L3 Switch provides HQ SVI gateways/inter-VLAN routing; branch Cisco router uses 802.1Q sub-interfaces
WAN	Use of the /30 links only for external transit
VPN	IKE/IPsec site-to-site between HQ-Router and BR-Router; services restrictions for 10.10.0.0/16 and 10.20.0.0/16
Virtualization	Proxmox, 1 dedicated port connected to HQ-L2-SW01
File/NTP	Ubuntu Samba AD member and Chrony server on HQ-FILE01
Mail	Exchange Server 2019, internal email
Web	NGINX reverse proxy to a small internal service portal
Monitoring	LibreNMS with SNMPv3 preferred and for system logs
Inventory	Snipe-IT
Security	Management isolation, secure protocols, disabled all unused ports, backups, endpoint/server firewalls, authorized vulnerability assessment

1.6 User Roles

Account / Role	Purpose	Key restrictions
Standard domain user	Email, file access, intranet, normal workstation tasks	No local/server/network-device administrative rights
Department security group	Controls access to departmental resources	Membership follows job role and manager approval
IT admin account	Infrastructure/server administration	Used only for administrative tasks; MFA where supported; not used for email/web browsing
Network admin account	Cisco/Fortinet/Proxmox device management	Allowed only from management VLAN or approved admin workstation
Service account	Application or scheduled service identity	No interactive logon unless specifically required; least privilege
Backup operator	Backup media rotation and restore tests	Access limited to backup workflow and protected media

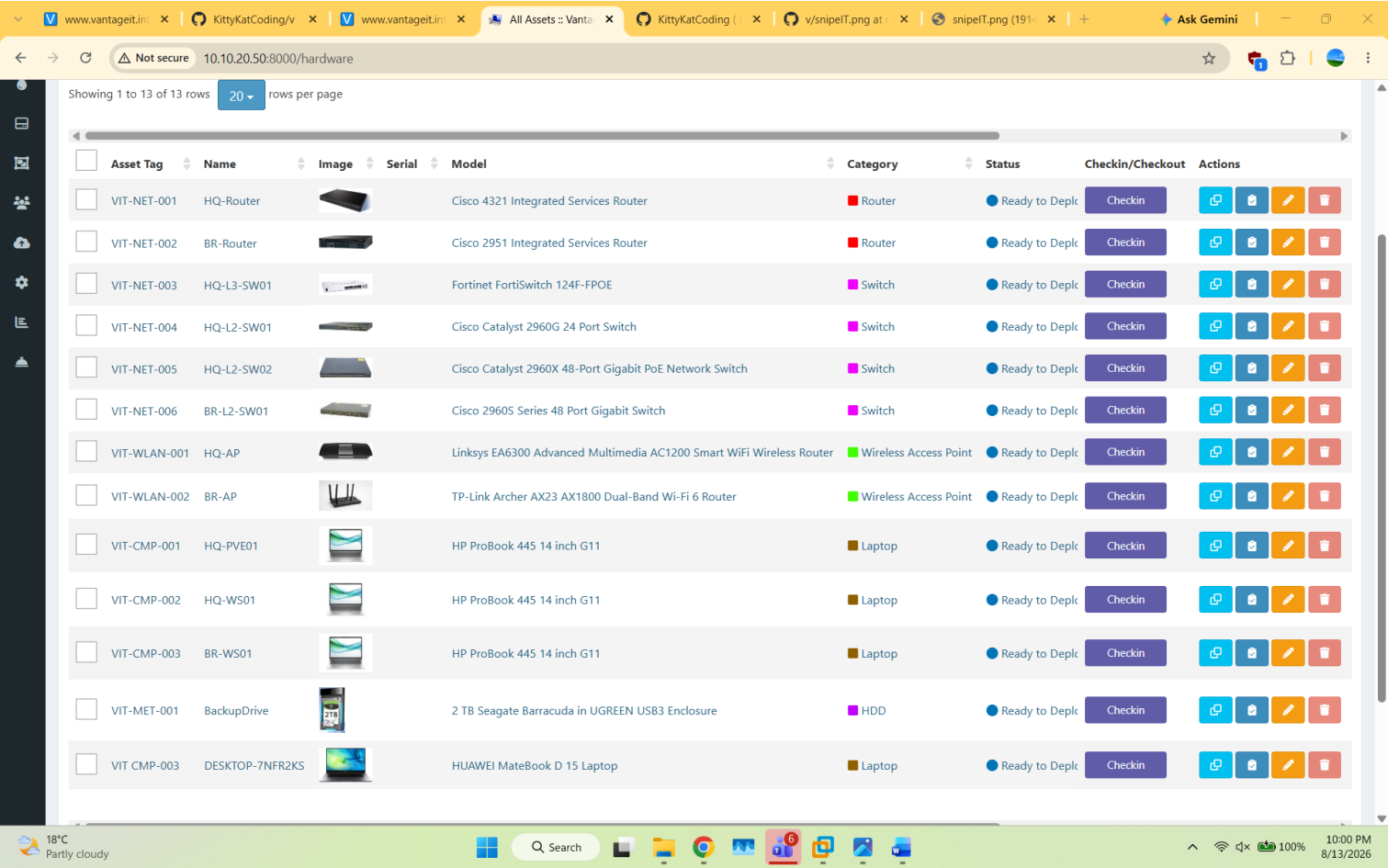
1.7 Naming Conventions

Object	Standard	Examples
Sites	HQ, BR	HQ-Router, BR-Router
Network devices	<SITE>-<ROLE>-<NN>	HQ-L3-SW01, HQ-L2-SW02, BR-L2-SW01
Servers	HQ-<SERVICE>-<NN>	HQ-DC01, HQ-FILE01, HQ-NMS01
Workstations	<SITE>-WS<NN>	HQ-WS01, BR-WS01
Access points	<SITE>-AP	HQ-AP, BR-AP
Users	first.last	alex.lee
Admin users	adm-first.last	evan.admin (sample format)
Service accounts	svc-<service>	svc-backup
AD groups	GG-<purpose>-<access>	GG-File-Projects-RW
Domain	vantageit.local	Internal AD/DNS namespace
Email	first.last@vantageit.local	Internal-only mail domain

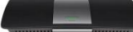
2. Equipment Inventory

Asset Tag	Item	Type	Qty	Site	Hostname / Label	Usage
VIT-NET-001	Cisco ISR 4321	Router	1	HQ	HQ-Router	WAN edge and VPN endpoint
VIT-NET-002	Cisco 2900 Series	Router	1	Branch	BR-Router	WAN edge, branch VLAN gateways, VPN endpoint
VIT-NET-003	FortiSwitch 124F-FPOE	Layer 3 switch	1	HQ	HQ-L3-SW01	HQ core, SVI gateways, inter-VLAN routing
VIT-NET-004	Cisco Catalyst 2960	Layer 2 Switch	1	HQ	HQ-L2-SW01	Proxmox + wired access
VIT-NET-005	Cisco Catalyst 2960	Layer 2 Switch	1	HQ	HQ-L2-SW02	HQ users + access point
VIT-NET-006	Cisco Catalyst 2960	Layer 2 Switch	1	Branch	BR-L2-SW01	Branch wired/AP access
VIT-WLAN-001	Linksys EA6300	Wireless Access Point	1	HQ	HQ-AP	Wireless access only
VIT-WLAN-002	TP-Link Archer AX23	Wireless Access Point	1	Branch	BR-AP	Wireless access only
VIT-CMP-001	HP ProBook, 32 GB RAM, 1 TB SSD, Core Ultra 5	Laptop/virtualization host	1	HQ	HQ-PVE01	Proxmox host (VE)
VIT-CMP-002	HP ProBook	Laptop/workstation	1	HQ	HQ-WS01	HQ domain workstation
VIT-CMP-003	HP ProBook	Laptop/workstation	1	Branch	BR-WS01	Branch domain workstation
VIT-MED-001/2	USB flash drive	Backup media	2	HQ	Backup-A / Backup-B	Encrypted offline backup rotation
VIT-CAB-(001 to 014)	Cat6 patch cable	Cables	14	Both	Patch cables	Device-to-patch-panel / rack patching

- Assets records are logged and maintained through Snipe-IT

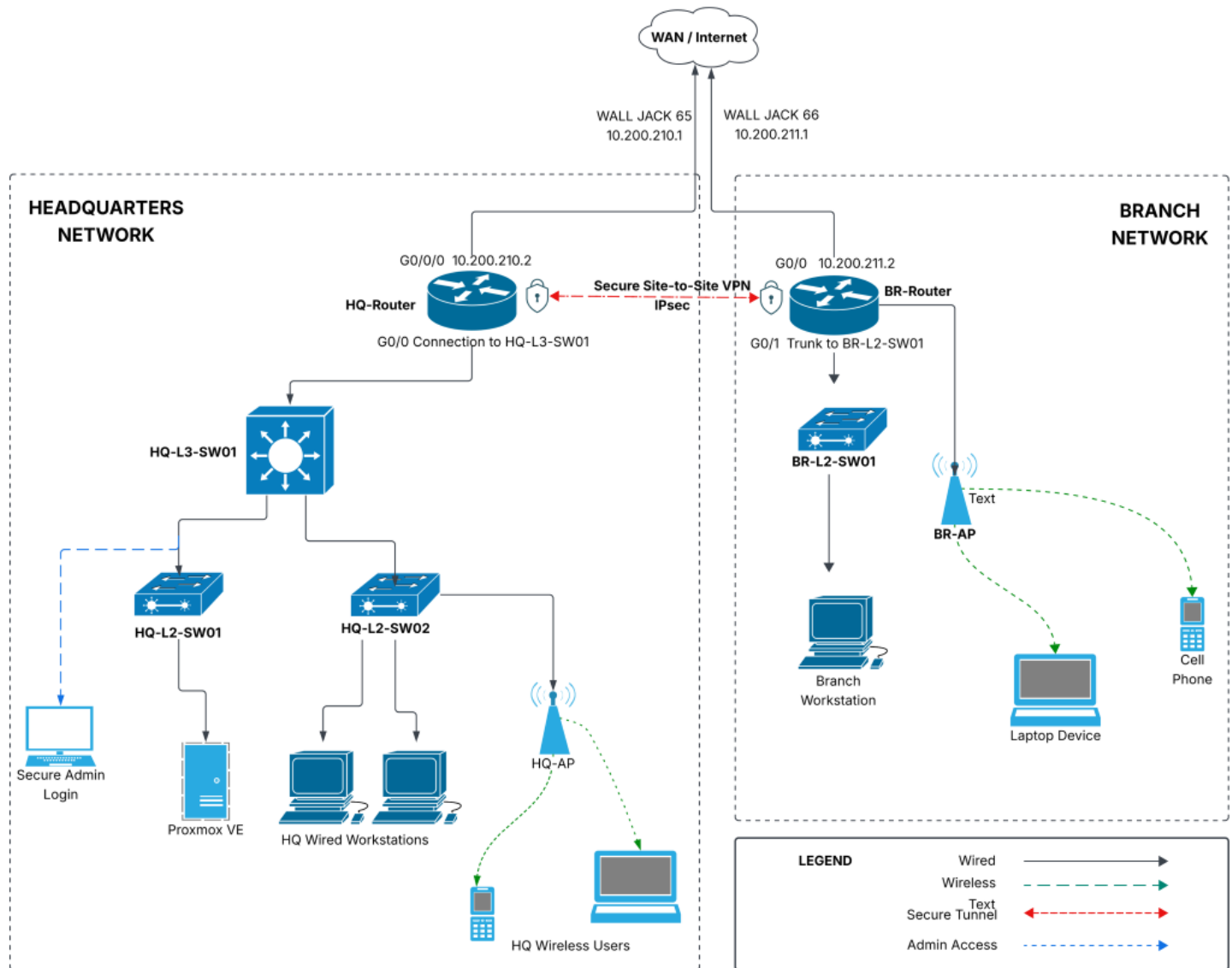


Screenshot from snipeit.vantageit.local

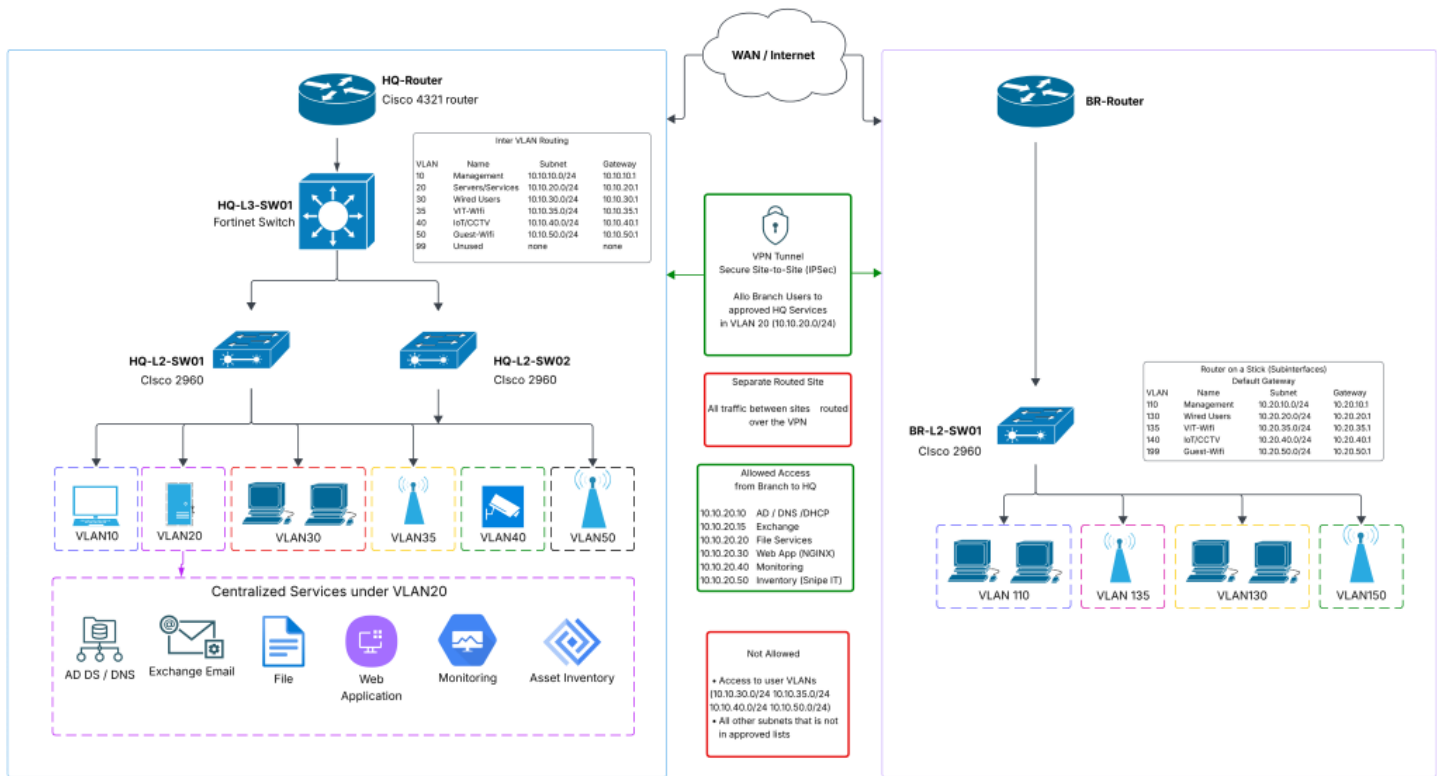
Asset Tag	Name	Image	Model	Category	Location
VIT-NET-001	HQ-Router		Cisco 4321 Integrated Services Router	Router	HQ
VIT-NET-003	HQ-L3-SW01		Fortinet FortiSwitch 124F-FPOE	Switch	HQ
VIT-NET-004	HQ-L2-SW01		Cisco Catalyst 2960G 24 Port Switch	Switch	HQ
VIT-NET-005	HQ-L2-SW02		Cisco Catalyst 2960X 48-Port Gigabit PoE Network Switch	Switch	HQ
VIT-WLAN-001	HQ-AP		Linksys EA6300 Advanced Multimedia AC1200 Smart WiFi Wireless Router	Wireless Access Point	HQ
VIT-CMP-001	HQ-PVE01		HP ProBook 445 14 inch G11	Laptop	HQ
VIT-CMP-002	HQ-WS01		HP ProBook 445 14 inch G11	Laptop	HQ
VIT-MET-001	BackupDrive		2 TB Seagate Barracuda in UGREEN USB3 Enclosure	HDD	HQ
VIT-NET-002	BR-Router		Cisco 2951 Integrated Services Router	Router	Branch
VIT-NET-006	BR-L2-SW01		Cisco 2960S Series 48 Port Gigabit Switch	Switch	Branch
VIT-WLAN-002	BR-AP		TP-Link Archer AX23 AX1800 Dual-Band Wi-Fi 6 Router	Wireless Access Point	Branch
VIT-CMP-003	BR-WS01		HP ProBook 445 14 inch G11	Laptop	Branch
VIT CMP-003	DESKTOP-7NFR2KS		HUAWEI MateBook D 15 Laptop	Laptop	Branch

3. Final Physical and Logical Topologies

3.1 Physical Topology

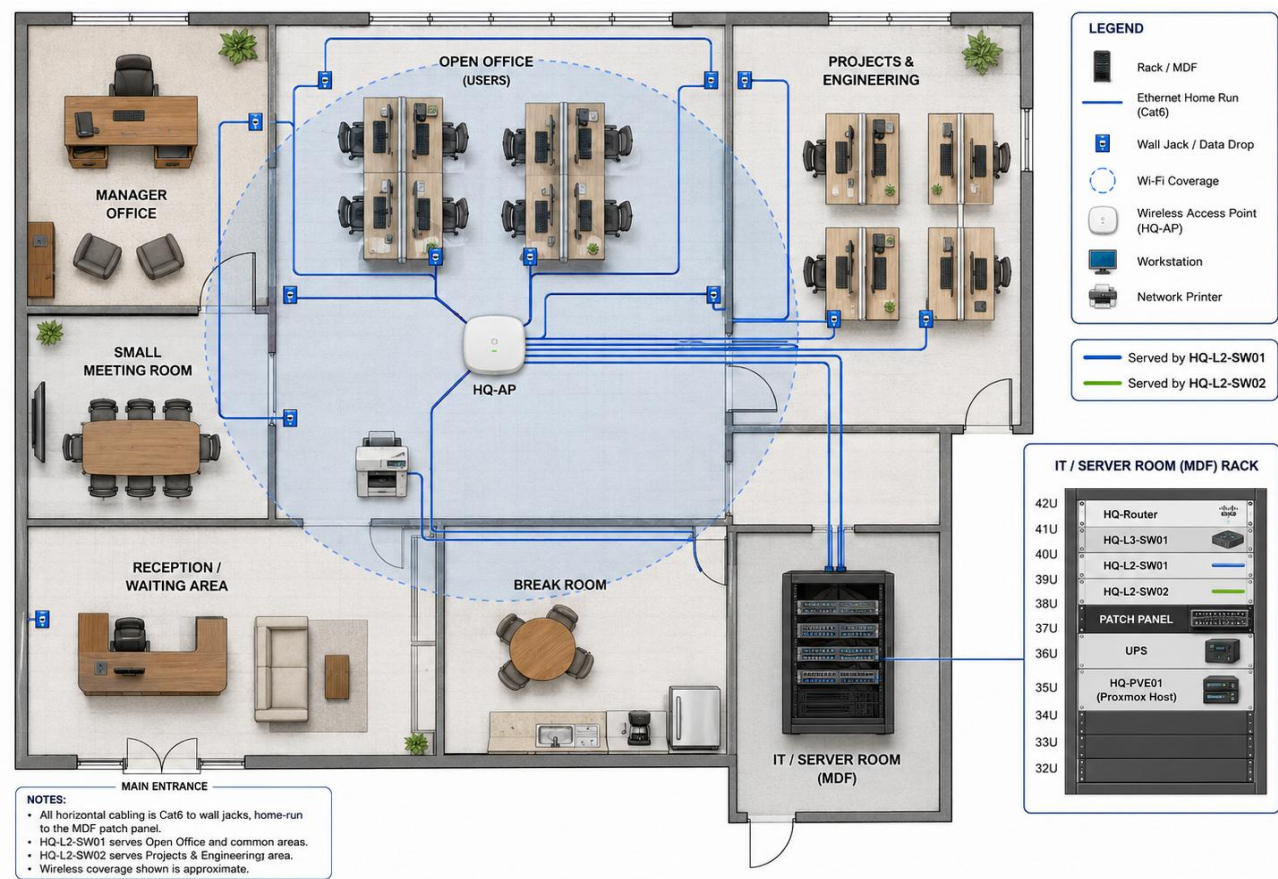


3.2 Logical Topology



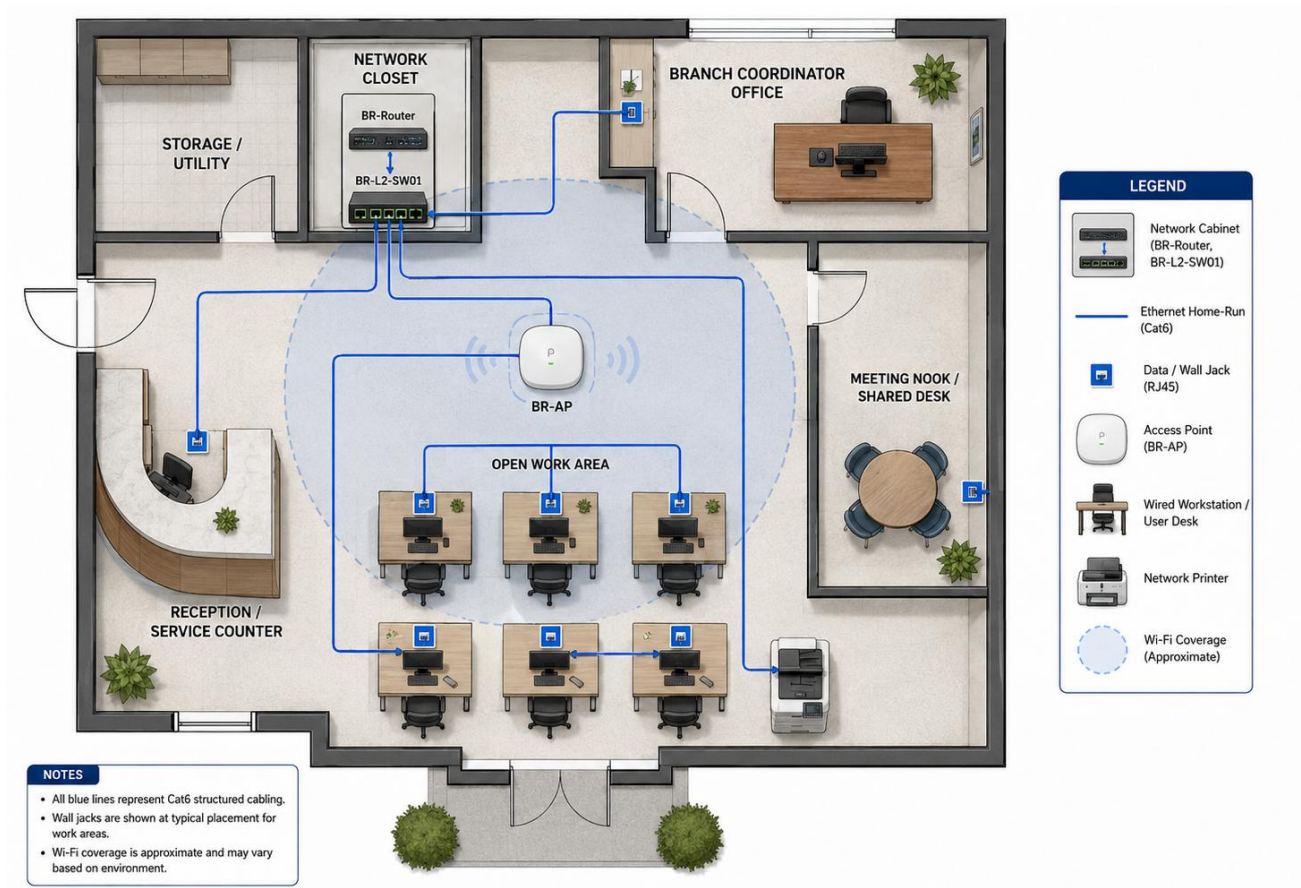
4. Headquarters and Brach Design

41. Headquarters Floor Plan



Drop Range	Room / Purpose	Recommended Access Switch / VLAN
HQ-D01-02	Reception area	HQ-L2-SW02 / VLAN30
HQ-D03-10	Open office (wired users)	HQ-L2-SW02 / VLAN30
HQ-D11-14	Management/finance	HQ-L2-SW01 / VLAN30
HQ-D15-20	Projects/engineering	HQ-L2-SW01 / VLAN30
HQ-D21	Meeting-room AP	HQ-L2-SW02 / VLAN35 corporate WLAN
HQ-D22-23	Printer/storage	HQ-L2-SW02 / VLAN30; future (recommendation)
HQ-D24-27	Spare/growth	Administratively disabled until assigned

4.2 Branch Floor Plan



Drop Range	Room / Purpose	Recommended VLAN
BR-D01-02	Reception/coordinator	VLAN130
BR-D03-08	Branch open office	VLAN130
BR-D09-10	Meeting room	VLAN130
BR-D11-14	Service desk	VLAN130
BR-D15-16	Printer/break area	VLAN130 or future isolated device VLAN
BR-D17-20	Growth/spare	Disabled/parking VLAN199 until assigned

5. WAN and VPN

5.1 Assigned WAN

Site	Wall Jack	Router WAN	Prefix	Gateway
Headquarters	65	10.200.210.2	/30	10.200.210.1
Branch	66	10.200.211.2	/30	10.200.211.1

5.2 Headquarters Route Plan

- HQ-L3-SW01 default route: 0.0.0.0/0 to 10.10.254.1 (HQ-Router).
- HQ-Router static route: 10.10.0.0/16 to 10.10.254.2 (HQ-L3-SW01).
- HQ-Router default route: 0.0.0.0/0 to 10.200.210.1
- Branch 10.20.0.0/16 is reachable through the VPN

5.3 Branch Route Plan

- BR-Router is directly connected to VLAN110, VLAN130, VLAN135, VLAN150, and VLAN199 through router-on-a-stick sub interfaces
- BR-Router default route: 0.0.0.0/0 to 10.200.211.1
- HQ summary 10.10.0.0/16 is protected by the VPN; service ACLs permit only approved traffic

5.4 Site to site VPN

Parameter	Design
VPN type	Cisco IOS policy-based IKE/IPsec site-to-site tunnel mode
Peers	HQ 10.200.210.2 to BR 10.200.211.2
Protected networks	HQ 10.10.0.0/16 and BR 10.20.0.0/16, restrict with ACLs
IKE authentication	Pre-shared key (secret key not shown)
IKE encryption	AES-256
IKE hash	SHA-256
DH group	Group 14
IPsec	ESP AES-256 + integrity; PFS group 14 where
Routing	Static routing appropriate for a two-site network

5.4 Router Basic and VPN Configuration

Configure HQ-Router:

1. Connect by console, erase current configuration, reload, and answer No to the initial configuration dialog. Confirm the interface names with show ip interface brief.
2. Set hostname HQ-Router, disable DNS lookup for mistyped commands, configure a local privileged administrator with a secret, generate RSA keys, enable SSH version 2, and restrict VTY access to the management source network. Do not document the real secret.
3. Configure the WAN interface with 10.200.210.2/30 and no shutdown. Add the default route to 10.200.210.1.
4. Configure the inside transit interface with 10.10.254.1/30 and no shutdown. Add ip route 10.10.0.0 255.255.0.0 10.10.254.2.
5. Verify local routing before creating the VPN: ping 10.10.254.2 and the upstream gateway. A failure here must be fixed before IPsec troubleshooting.
6. Create an extended crypto ACL matching HQ 10.10.0.0/16 to branch 10.20.0.0/16. This defines encryption interest; separate service ACLs enforce least privilege.
7. Create an IKE policy using mutually supported AES-256, SHA-256 (or the strongest supported equivalent), pre-shared authentication, and DH group 14. Configure the Branch WAN peer with a pre-shared key.
8. Create the IPsec transform set, crypto map, peer, and PFS option. Apply the crypto map to the HQ WAN interface.
9. Generate interesting traffic from HQ to a permitted branch address, then verify show crypto isakmp sa / show crypto ikev2 sa as appropriate and show crypto ipsec sa.

```
hostname HQ-Router
no ip domain-lookup
ip domain-name vantageit.local
enable secret <HIDDEN-NOT-SHOWN>
service password-encryption
username admin secret <HIDDEN-NOT-SHOWN>
!
interface GigabitEthernet0/0/0
description connection to WALL-JACK65
ip address 10.200.210.2 255.255.255.252
ip nat outside
negotiation auto
crypto map VIT-MAP
!
interface GigabitEthernet0/0/1
description TRUNK-TO-HQ-L3-SW01
ip address 10.10.254.1 255.255.255.252
ip nat inside
negotiation auto
!
interface GigabitEthernet0/0/1.10
description VLAN10-MGMT
encapsulation dot1Q 10
ip address 10.10.10.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.20
description VLAN20-SERVERS
encapsulation dot1Q 20
```

```
ip address 10.10.20.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.30
description VLAN30-STAFF
encapsulation dot1Q 30
ip address 10.10.30.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.35
description VLAN35-VIT-WIFI
encapsulation dot1Q 35
ip address 10.10.35.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.40
description VLAN40-IoT-CCTV
encapsulation dot1Q 40
ip address 10.10.40.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.50
description VLAN50-GUEST-WIFI
encapsulation dot1Q 50
ip address 10.10.50.1 255.255.255.0
ip nat inside
!
interface GigabitEthernet0/0/1.99
description VLAN99-NATIVE-UNUSED
ip route 10.10.0.0 255.255.0.0 10.10.254.2
ip route 0.0.0.0 0.0.0.0 10.200.210.1

ip access-list extended VPN-HQ-BR
permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
!
crypto isakmp policy 10
encr aes 256
hash sha256
authentication pre-share
group 14
!
crypto isakmp key <HIDDEN-NOT-SHOWN> address 10.200.211.2
crypto ipsec transform-set VIT-SET esp-aes 256 esp-sha-hmac
mode tunnel
!
crypto map VIT-MAP 10 ipsec-isakmp
set peer 10.200.211.2
set transform-set VIT-SET
set pfs group14
match address VPN-HQ-BR
!
interface GigabitEthernet0/0/0
crypto map VIT-MAP
```

6. VLAN and IP-Addressing Table

6.1 Headquarters

VLAN	Name	Subnet	Gateway	Addressing	Purpose
10	MGMT	10.10.10.0/24	10.10.10.1	Static	Management for Network Devices
sh20	SERVERS	10.10.20.0/24	10.10.20.1	Static	Centralized Servers/VMs
30	STAFF	10.10.30.0/24	10.10.30.1	DHCP	Wired Users/Printers
35	VIT-WIFI	10.10.35.0/24	10.10.35.1	DHCP	HQ-User Wifi access
40	IoT-CCTV	10.10.40.0/24	10.10.40.1	DHCP	Future recommendation
50	GUEST-WIFI	10.10.50.0/24	10.10.50.1	DHCP	Reserved guest subnet; Internet only when VLAN-capable AP exists
99	NATIVE-UNUSED	No client subnet	None	None	Native VLANs / Unused ports

6.2 Branch VLANs

VLAN	Name	Subnet	Gateway	Addressing	Purpose
110	MGMT	10.20.10.0/24	10.20.10.1	Static	Branch network management
130	WIRED-STAFF	10.20.30.0/24	10.20.30.1	DHCP via relay	Branch wired users/printer
135	CORP-WIFI	10.20.35.0/24	10.20.35.1	DHCP via relay	Branch corporate wireless
150	GUEST-WIFI	10.20.50.0/24	10.20.50.1	DHCP via relay/local alternative	Internet-only guest network if hardware supports
199	NATIVE-UNUSED	None	None	None	Native/Unused

6.3 DHCP Scope

Scope	Range	Router	DNS	Notes
HQ-STAFF-V30	10.10.30.50-220	10.10.30.1	10.10.20.10	.1-.49 reserved .221-.254 spare
HQ-WIFI-V35	10.10.35.50-220	10.10.35.1	10.10.20.10	HQ wireless
HQ-IOT-V40	10.10.40.50-200	10.10.40.1	10.10.20.10	For future use
HQ-GUEST-V50	10.10.50.50-230	10.10.50.1	10.10.20.10	Internet-only
BR-WIRED-V130	10.20.30.50-220	10.20.30.1	10.10.20.10	Relayed across VPN
BR-WIFI-V135	10.20.35.50-220	10.20.35.1	10.10.20.10	Relayed across VPN
BR-GUEST-V150	10.20.50.50-230	10.20.50.1	10.20.50.1	Prefer local DHCP

7. Device and Port Assignment

7.1 Headquarters Core Switch Access (HQ-L3-SW101)

HQ-L3-SW01 Port	Connected Device	Mode	Native/Untagged	Tagged/Allowed	Notes
port1	HQ-L2-SW01 uplink	Trunk	99	10,20,30,35,40,50	Restrict list; RSTP trunk
port2	HQ-L2-SW02 uplink	Trunk	99	10,20,30,35,40,50	Restrict list; RSTP trunk
port23/24 (example)	HQ-Router transit	Routed interface or dedicated VLAN	-	10.10.254.0/30	Use routed-port method
Unused	None	Disabled/access parking	99	None	Administratively disabled

7.2 Headquarters Access Switch 1 (HQ-L2-SW01)

Port	Mode / VLAN	Connected Device / Use	Security
Gi0/1	Trunk; native 99; allow 10,20,30,35,40,50	Connection to HQ-L3-SW01	Fixed trunk restricted VLAN list
Gi0/10	Trunk; native 99; allow 10,20,30	HQ-PVE01	Fixed trunk
Gi0/2	Access VLAN35	VIT-WIFI	PortFast, BPDU Guard, port security
Gi0/3-9,11-23	Access VLAN30	HQ workstations and printers	PortFast, BPDU Guard, port security
Gi0/24	Management	IT admin access	IT admin local logging

7.3 Headquarters Access Switch 2 (HQ-L2-SW02)

Port	Mode / VLAN	Connected Device / Use	Security
Gi1/0/1	Trunk; native 99; allow 10,20,30,35,40,50	Uplink to HQ-L3-SW01	Fixed trunk; restricted VLANs
Gi1/0/1	Access VLAN30	HQ-WS01	PortFast, BPDU Guard
Gi1/0/2	Access VLAN35	HQ-AP (Vantage users WLAN)	PortFast; one AP uplink
Gi1/0/3-12	Access VLAN30	Reception/service desk/printer	PortFast, BPDU Guard, port security
Gi1/0/13-24	Access VLAN99 + shutdown	Unused	Disabled
Gi1/0/48	Management	IT admin access	IT admin local logging

7.4 Branch Switch (BR-L2-SW01)

Port	Mode / VLAN	Connected Device / Use	Security
Gi1/0/1	802.1Q trunk; native 199; allow 110,130,135,150,199	BR-Router	Fixed trunk; restricted VLAN list
Gi1/0/48	Access VLAN130	IT admin laptop	PortFast, BPDU Guard, port security
Gi1/0/2	Access VLAN135	BR-AP	PortFast; AP mode
Gi1/0/3-24	Access VLAN130	Branch staff/printer drops	PortFast, BPDU Guard, port security
Gi1/0/25-47	Access VLAN199 and shutdown	Unused	Disabled

7.5 Layer 2 Switches Configuration Procedure

1. Console to the switch, erase any previous configuration only if authorized, and reload.
2. Set the assigned hostname and management domain name. Create a unique local administrator secret. Generate RSA keys, enable SSH version 2, and restrict VTY access to the management subnet.
3. Create only the VLANs required at that site and name them exactly as documented.
4. Configure the management SVI in the management VLAN and ip default-gateway to the local VLAN gateway.
5. Configure uplinks as static trunks with an explicit native/unused VLAN and explicit allowed-VLAN list. Avoid dynamic trunk negotiation.
6. Configure endpoint ports as access ports. Enable PortFast and BPDU Guard only on edge/access ports, never blindly on inter-switch trunks.
7. Apply port security to appropriate user ports. Use a small maximum MAC count and shutdown/restrict violation behavior based on the requirement.
8. Move all unused ports to the parking/native-unused VLAN and shut them down. Add descriptions to every used uplink and important endpoint port.
9. Save the configuration and capture show vlan brief, show interfaces trunk, show spanning-tree, show port-security, and show running-config.

```

configure terminal
hostname HQ-L2-SW01
no ip domain-lookup
ip domain-name vantageit.local
vlan 10
  name MGMT
vlan 20
  name SERVERS
vlan 30
  name STAFF
vlan 35
  name VIT-WIFI
vlan 40
  name IOT-CCTV
vlan 50
  name GUEST-WIFI
vlan 99
  name NATIVE-UNUSED
!
interface GigabitEthernet0/1
  description TRUNK-TO-HQ-L3-SW01
  switchport mode trunk
  switchport trunk native vlan 99
  switchport trunk allowed vlan 10,20,30,35,40,50,99
  switchport nonegotiate
!
interface GigabitEthernet0/10
  description TRUNK-TO-HQ-PVE01

```

```
switchport mode trunk
switchport trunk native vlan 99
switchport trunk allowed vlan 10,20,30,99
switchport nonegotiate
!
interface range GigabitEthernet0/2
switchport mode access
switchport access vlan 35
spanning-tree portfast
spanning-tree bpduguard enable

!
interface range GigabitEthernet0/3-9
switchport mode access
switchport access vlan 30
spanning-tree portfast
spanning-tree bpduguard enable
shutdown
!
interface Vlan10
ip address 10.10.10.3 255.255.255.0
no shutdown
ip default-gateway 10.10.10.1
```

8. Virtualization

8.1 Virtual Environment Selection

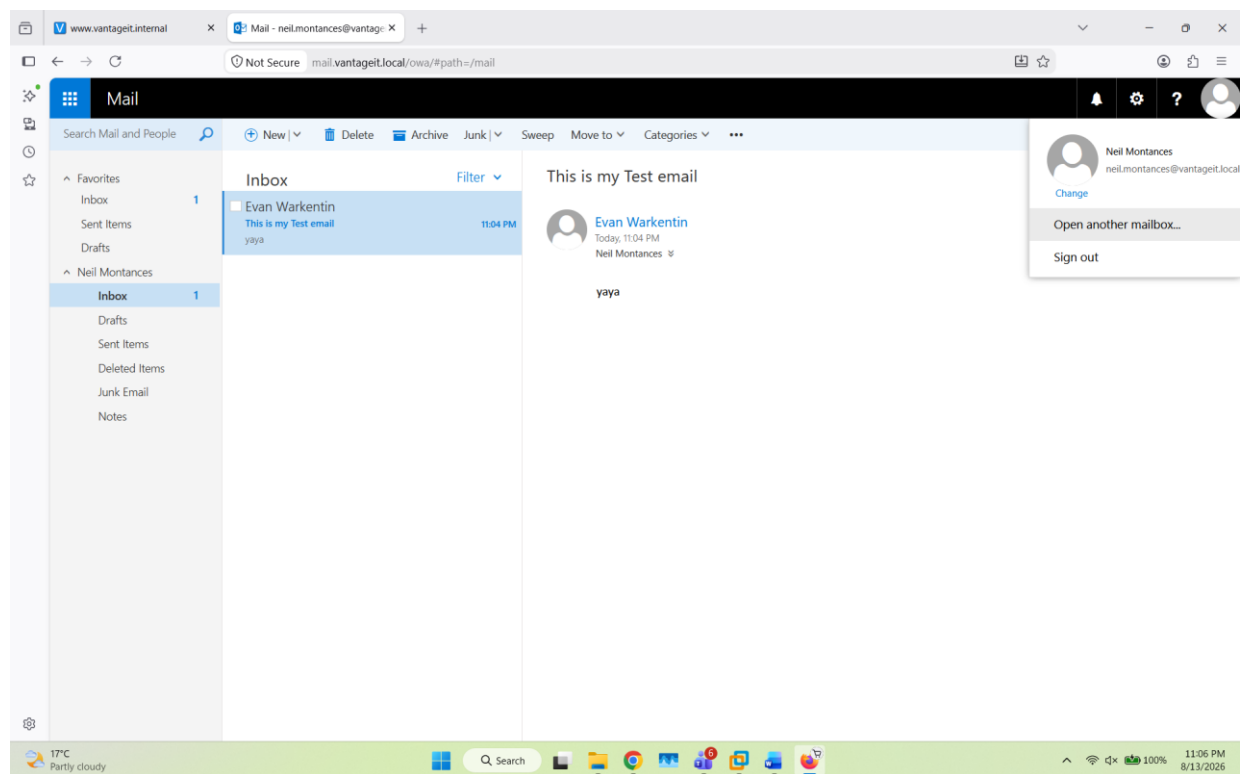
Proxmox Server virtual environment was selected because it does support both KVM virtual machines and Linux containers in one management interface, VLAN-aware Linux bridges, scheduled backup jobs, console access, and centralized administration. The assigned HP ProBook uses 32 GB of RAM and 1 TB of storage. This is suitable to allocate the resources and select only lightweight services for non essential functions.

Guest	Type	vCPU	RAM	Disk	IP	Role
VM100 HQ-DC01	Windows VM	2	4 GB	80 GB	10.10.20.10	AD DS/DNS/DHCP
VM110 HQ-EX01	Windows VM	4	10 GB	160 GB	10.10.20.15	Exchange 2019
VM120 HQ-FILE01	Ubuntu VM	2	3 GB	180 GB	10.10.20.20	Samba/Chrony/SNMP
CT130 HQ-WEB01	Ubuntu/Debian CT	1	1 GB	30 GB	10.10.20.30	NGINX + web app
CT140 HQ-NMS01	Ubuntu/Debian CT	2	2 GB	45 GB	10.10.20.40	LibreNMS
CT150 HQ-ASSET01	Ubuntu/Debian CT	2	2 GB	45 GB	10.10.20.50	Snipe-IT

9. Servers and Services

9.1 Service Map

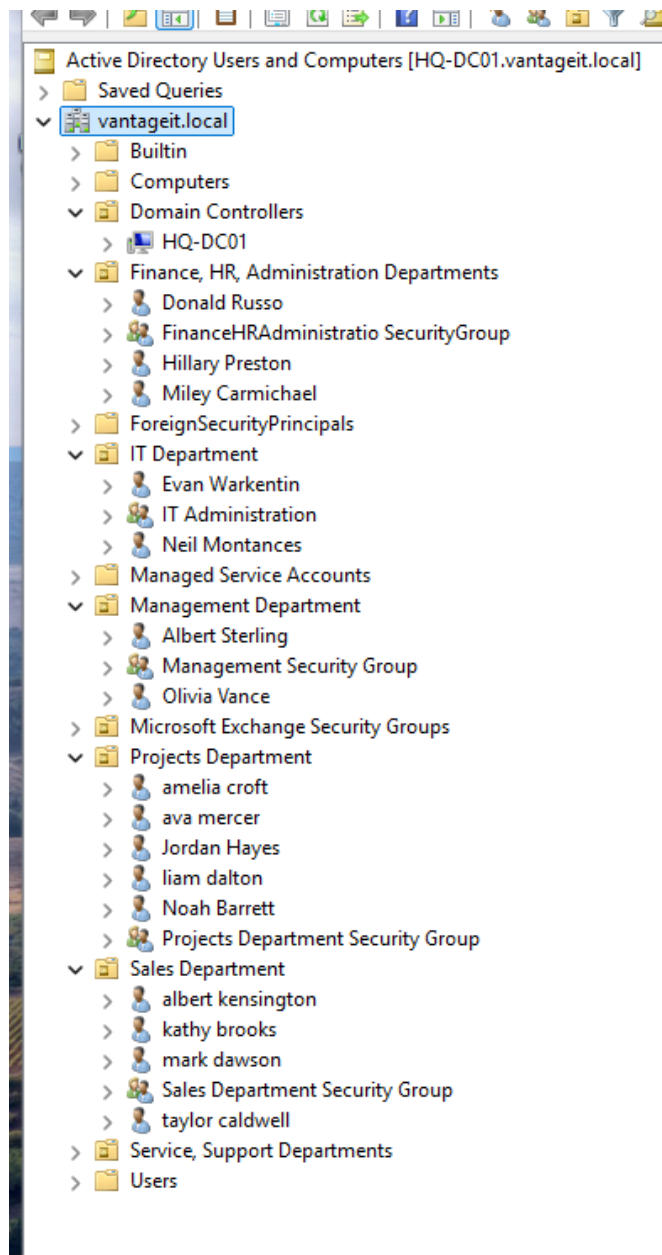
Hostname	Platform	IP	Primary Services	Authorized Users
HQ-DC01	Windows Server 2025	10.10.20.10	ADDS, DNS	All domain clients; IT admin for management
HQ-EX01	Windows Server 2025	10.10.20.15	Exchange Server 2019 internal mail	HQ/BR domain users
HQ-FILE01	Ubuntu Server	10.10.20.20	Samba AD member, Chrony NTP, SNMP agent	Department groups; IT admin
HQ-WEB01	Ubuntu Server	10.10.20.30	NGINX + Unicorn/Flask + SQLite internal app	HQ/BR users
HQ-NMS01	Ubuntu Server	10.10.20.40	LibreNMS	IT admin
HQ-ASSET01	Ubuntu Server	10.10.20.50	Snipe-IT	IT admin



We have local emails using an exchange server with <https://mail.vantageit.local> users login with their domain name and password.

9. Active Directory Domain Services (HQ-DC01)

We have an active directory domain service with an ip of 10.10.20.10 and domain vantageit.local



Management Department

Albert.Sterling@vantageit.local

Olivia.Vance@vantageit.local

Sales Department

Kathy.Brooks@vantageit.local

Mark.Dawson@vantageit.local

Taylor.Caldwell@vantageit.local

Albert.Kensington@vantageit.local

Projects Department

Noah.Barrett@vantageit.local

Jordan.Hayes@vantageit.local

Amelia.Croft@vantageit.local

Ava.Mercer@vantageit.local

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Service and Support Department

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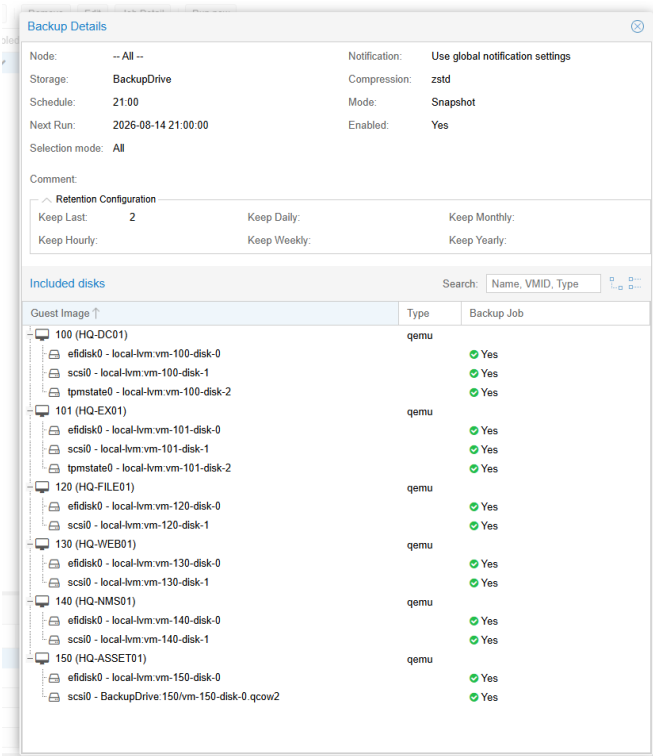
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10. Security

We have secure passwords for all logins on servers, switches, and routers. All servers have the latest security updates. We also have password encryption on the switches and SSH.

11. Backup and Recovery



The whole Proxmox server is backed up every day at 9 pm to a 2 TB external SATA HDD in a UGREEN USB 3 enclosure. Retention is kept last because I have 2 TB to work with.

12. Monitoring or Logging

Monitoring was used to answer three common questions:

- Is the device/service reachable? Is its health changing in a way that could become a failure?
- Is there useful evidence to explain what happened?
- LibreNMS provides the central infrastructure view, while Windows Event Viewer, Linux journal/application logs, Proxmox task logs, and device logs provide deeper event evidence

All servers are monitored and logged in the librenms sever:

The screenshot displays the LibreNMS web interface in a browser window. The address bar shows the URL `librenms.vantageit.local:8000/devices`. The interface includes a navigation menu with options like Overview, Devices, Maps, Ports, Health, Services, Apps, and Alerts. The main content area shows a list of devices with columns for S. (Status), Id, M. (Metric), Vendor, Device, Metrics, Platform, Device OS, Up/Down Time, Location, and Actions. The devices listed are:

S.	Id	M.	Vendor	Device	Metrics	Platform	Device OS	Up/Down Time	Location	Actions
2	HQ-DC01	10.10.20.10	AMD x64	Microsoft Windows Server 2025 Datacenter (24H2) (Multiprocessor)	4	AMD x64	Microsoft Windows Server 2025 (24H2) (Multiprocessor)	10h 5m 37s		
5	HQ-EX01	10.10.20.15	AMD x64	Microsoft Windows Server 2025 (24H2) (Multiprocessor)	16	AMD x64	Microsoft Windows Server 2025 (24H2) (Multiprocessor)	7h 35m 38s		
8	hq-file01	10.10.20.20	Generic x86 64-bit	Linux 7.0.0-29-generic (Ubuntu 26.04)	2	Generic x86 64-bit	Linux 7.0.0-29-generic (Ubuntu 26.04)	2d 3h 42m	Server Rack 1	
9	hq-web01	10.10.20.30	Generic x86 64-bit	Linux 7.0.0-29-generic	2	Generic x86 64-bit	Linux 7.0.0-29-generic	2d 4h 27m	Unknown	
3	Internet	8.8.8.8	Ping only							
6	Proxmox	10.10.20.11	Generic x86 64-bit	proxmox pve 7.0.14-8-pve	27	Generic x86 64-bit	proxmox pve 7.0.14-8-pve	2d 10h 6m	Sitting on the Dock of the Bay	

The interface also shows a filter bar, a search bar, and a pagination bar at the bottom indicating "Showing 1 to 6 of 6 entries".

<http://librenms.vantageit.local:8000/>

13. Testing and Troubleshooting

We did a lot of testing and troubleshooting, to troubleshoot internet access we first ping default gateway to see if the router is working, then ping 8.8.8.8 to see if internet is active.

14. Risk Register

Risk register of the Proxmox Laptop

Risk	Risk Description	1-5 likely	1-5 impact	Risk Score (L x I)	Proactive Mitigation Strategy	Reactive Contingency Plan
1	Hardware Failure	2	5	10	Check thermals on all devices	Buy new hardware if it fails
2	Power Interruption	4	5	20	Install an uninterruptable Power supply	Restart all devices and recover from backups if data is lost
3	Storage Failure	1	5	5	check ssd and hdd health data	if the ssd fails, restore proxmox from hdd backup onto new ssd. If backup hdd fails, buy and install new hdd
4	Network Instability	3	3	9	connect everything well with good configs	inspect cables and use pings and tracerouts to see what is wrong
5	Thermal Throttling	2	1	2	Put server laptop onn stand to cool off	place a box fan pointed at the laptop
6	Accidental Disturbance	2	2	4	Make sure no one touches it or I touch it	fix everything that changed

15. Work Logs

Date	Day	Work completed / documented
Aug 4, 2026	Tuesday	Planning and brainstorming; Company creation, review the requirements for Vantage IT Solutions, departments, roles, naming standards, two-site architecture and security boundaries. Received workstation devices, inspect and test, download needed OS. Wipe the device, install needed software's
Aug 5, 2026	Wednesday	Received network devices, check, test and wipe current configurations; Designed HQ/Branch VLANs, IP addressing, WAN/transit plan, device hostnames, Device assignment
Aug 6, 2026	Thursday	Created physical/logical topology HQ core/access and branch router-on-a-stick design.
Aug 7, 2026	Friday	Prepared Cisco/FortiSwitch configuration procedures, wireless AP-mode procedures, routing and site-to-site VPN plan.
Aug 10, 2026	Monday	Proxmox installation, VM/container resource plan, AD DS/DNS/DHCP configuration and workstation domain-join procedures.
Aug 11, 2026	Tuesday	Prepared Ubuntu Samba/SNMP, NGINX internal application, Exchange lab procedure, and cross-site service-access plan.
Aug 12, 2026	Wednesday	Prepared LibreNMS and Snipe-IT procedures, security controls, vulnerability assessment plan, backup/recovery flow, and risk register.
Aug 13, 2026	Thursday	Completed validation/troubleshooting plan, configuration, screenshot evidence checklist, references, and final document review. Presentation prepared

Neil made most configurations to the switches and routers, Evan Made the VMS and active directory.

16. References

Publisher	Reference	Official URL
Fortinet	FortiSwitch documentation / Administration Guide	https://docs.fortinet.com/product/fortiswitch
Fortinet	FortiSwitch management ports / standalone administration	https://docs.fortinet.com/document/fortiswitch/6.4.2/administration-guide/790467/introduction
Cisco	Catalyst 2960 configuration and spanning-tree/PortFast/BPDU Guard documentation	https://www.cisco.com/c/en/us/support/switches/catalyst-2960-series-switches/series.html
Cisco	IOS site-to-site IPsec/IKE configuration examples	https://www.cisco.com/c/en/us/support/docs/ios-nx-os-software/ios/218432-configure-a-site-to-site-ipsec-ikev1-tun.html
Proxmox	Proxmox VE Administration Guide	https://pve.proxmox.com/pve-docs/pve-admin-guide.html
Microsoft	Install Active Directory Domain Services / Windows Server documentation	https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/deploy/install-active-directory-domain-services--level-100-
Microsoft	DNS and DHCP Server documentation	https://learn.microsoft.com/windows-server/networking/
Microsoft	Exchange Server supportability matrix	https://learn.microsoft.com/en-us/exchange/plan-and-deploy/supportability-matrix
Microsoft	Exchange Server 2019 and 2016 end-of-support roadmap	https://learn.microsoft.com/troubleshoot/exchange/administration/exchange-2019-2016-end-of-support
Ubuntu / Samba	Ubuntu Server Samba documentation and Samba project docs	https://ubuntu.com/server/docs/explanation/intro-to/samba/ ; https://www.samba.org/samba/docs/
NGINX	NGINX documentation	https://nginx.org/en/docs/
LibreNMS	LibreNMS installation and device documentation	https://docs.librenms.org/Installation/Install-LibreNMS/ ; https://docs.librenms.org/Support/Adding-a-Device/
Snipe-IT	Snipe-IT documentation	https://snipe-it.readme.io/
TP-Link	Archer AX23 support/downloads and user guide	https://www.tp-link.com/support/download/archer-ax23/
Linksys	EA6300 support and bridge/access-point guidance	https://support.linksys.com/
Diagrams	Lucid app (Trial version)	https://lucid.app/users/login#/login